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Cummins® - Branded Prestolite Starters, Alternators, Service, and Repair

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This Service Bulletin covers service intervals, service, repair kits, standard repair times, service tools, and literature to support Cummins® - Branded Prestolite Alternators and Starter motor, for the B,C,L,M,N,K, and Q Series engines.

Section 1. Service Intervals

It is recommended that the branded Starters and Alternators be maintained at the following service intervals:

- Starters:

20,000 strikes, 120,000 Miles [192,000 Km], 10,000 Hours, or 2 Years; whichever occurs first.

Note : Under circumstances where the number of strikes are not accumulated at a fast rate (e.g. stand-by gensets), perform service after 20,000 strikes, not hours of operation or years.

- Alternators:

120,000 Miles [192,000 Km], 10,000 Hours or 2 Years; whichever occurs first.

It is recommended that before starting a Starter motor or Alternator service or repair, reference be made to Section 7 in this service bulletin.

When performing Starter or Alternator service or repairs, general safety precautions **must** be followed to provide personal safety, as outlined in Cummins Inc. service manuals.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Acid is extremely dangerous and can damage the machinery and can also cause serious burns. Always provide a tank of strong soda water as a neutralizing agent when servicing the batteries. Wear goggles and protective clothing to reduce the possibility of serious personal injury.

⚠ CAUTION ⚠

To prevent damage to the Starter, do not engage the starting motor for more than 30 seconds at a time.

⚠ CAUTION ⚠

The Starter is an intermittent duty electric motor that is capable of high outputs for short periods of time. Excessive cranking of the starting motor will lead to a component malfunction.

Wait 2 minutes between each attempt to start (electrical starting motors **only**).

If the engine does **not** start after three attempts, check the fuel supply system.

Absence of blue or white exhaust smoke during cranking indicates that fuel is **not** being delivered. The starter **must not** be used to prime the fuel system. Reference the appropriate base engine Troubleshooting and Repair Manual for the correct engine specific priming procedure.

Section 2. Service and Repair Kits, and Part Number Reference

This section tabulates the Cummins® - Branded Alternators and Starter Motors, and their service and repair kits.

Cummins® Part Number	ReCon® Part Number	Prestolite™ Part Number	Component	Voltage	Prestolite™ Model Number	Cummins® Kit Part Number
213807	N/A	859468	Alternator	24-VDC	A001.2301 JB	3806193
216037	N/A	859469	Alternator	24-VDC	A001.2300 JB	3806193
218569	N/A	859495	Alternator	24-VDC	A001.2184 CA	3806194
3001774	3001774-RX	859472	Alternator	24-VDC	A001.2509 JB	3806193
3015516	3015516-RX	859473	Alternator	24-VDC	A001.2303 JB	3806193
3026259	3026259-RX	859475	Alternator	24-VDC	A001.4626 JC	3806190

3084227	3084227-RX	859476	Alternator	24-VDC	AMA.5104 UT	3806195
3088320	3088320-RX	859477	Alternator	24-VDC	8SC3009Z A[110-258]	3806191
3253477	3253477-RX	859480	Alternator	24-VDC	A4024/15[857606]	3806188
3254908	3254908-RX	859482	Alternator	24-VDC	A24R/12[857265]	3806189
3256759	N/A	859544	Alternator	24-VDC	LNA5524/7	3806192
3279007	3279007-RX	859487	Alternator	24-VDC	A5524/58[858693]	3806188
3279220	3279220-RX	859489	Alternator	24-VDC	8SC3014U [110-298]	3806191
3279416 EC 40 ¹	N/A	N/A	Alternator	24-VDC	N/A	N/A
3332215	N/A	859617	Alternator	24-VDC	LNA4024/7	3806192
3631678	N/A	859673	Alternator	24-VDC	LNA4024/8	3806192
185822	N/A	859467	Starter	24-VDC	M001.740 6MC	3806198
216718	N/A	859494	Starter	12-VDC	M001.728 8MC	3806198
707328	707328-RX	859470	Starter	24-VDC	MS1/101[858442]	3806196
707330	707330-RX	859471	Starter	24-VDC	MS1/100[858441]	3806196
3017122	N/A	859474	Starter	24-VDC	M001.094 134	3806199
3171479	N/A	859685	Starter	24-VDC	MS7/11	3806196
3253498	N/A	859481	Starter	24-VDC	MS1/107[858448]	3806196
3255652	3255652-RX	859483	Starter	24-VDC	MS1/126[859200]	3806196
3255829	3255829-RX	859484	Starter	24-VDC	MS1/127[859201]	3806196
3255868	3255868-RX	859485	Starter	24-VDC	MS1/129[858484]	3806196
3255945	3255945-RX	859486	Starter	24-VDC	MS1/130[859344]	3806196

3283565	3283565-RX	859499	Starter	24-VDC	MS3/51[857633]	3806196
3283586	3283586-RX	859500	Starter	24-VDC	LNS4524/6[859411]	3806197
3332270	N/A	859261	Starter	24-VDC	MS7/10	3806196
3632271	N/A	MS7-302C	Starter	24-VDC	MS7-302C	See Note ¹
3632273	N/A	MS7-301C	Starter	24-VDC	MS7-301C	See Note ¹
3632274	N/A	MS7-300C	Starter	24-VDC	MS7-300C	See Note ¹
3632272	N/A	MS7-312C	Starter	24-VDC	MS7-312C	See Note ¹
NOTE ¹ For additional information on superseding part number and Cummins® Kit Part Numbers, please contact Parts Marketing at Mechelen PDC, Belgium						

Section 3. Service and Repair Kits

Note : Individual components within the kits will not be made available.

The following Service and Repair Kits are used for the service and repair of the Cummins® - Branded Alternators and Starter Motors.

Cummins® Kit Part Number	Component Type	Kit Contents	Qty.	Comments:
3806196	Starter	Felt Wick	2	Reference Section 5 in this service bulletin (containing 091996) before commencing Starter motor service or repair.
SK27	MS1	Oil Seal (shift housing)	1	
	MS3	O-ring (comm. end & shift housing)	2	
	MS7	O-ring (nose housing)	1	
		Brush Assembly	8	
		Cam (lever arm)	2	
		Bushing (nose housing)	1	
		Oilite™ Brush (shift housing)	1	
		Thrust Washer (nose housing)	1	
		Boot (lever arm)	1	

3806188	Alternator	Brush Box Assembly	1	Reference Section 5 in this service bulletin (containing 091996) before commencing Alternator service or repair.
SK29	A3024	Bearing (drive end)	1	
	A4024	Bearing (slip ring end)	1	
	A5524	O-ring Seal	1	
		Woodruff Key	1	
		Slip Ring	1	
3806189	Alternator	Bearing (drive end)	1	Reference Section 5 in this service bulletin (containing 091996) before commencing Alternator service or repair.
SK30	A24	Bearing (slip ring end)	1	
		Woodruff Key	1	
		Diode Assembly	1	
		Brush Assembly	2	
		O-ring (slip ring end housing)	1	
		Sealing Ring (brush holder assembly)	1	
3806197	Starter	Bush (nose housing)	1	Reference Section 5 in this service bulletin (containing 091996) before commencing Starter motor service or repair.
SK31	LNS	O-ring (pivot bolt)	1	
		Bushing (shift housing)	1	
		Oil Seal (shift housing)	1	

		O-ring (-ve terminal)	1	When replacing -ve brush assembly, use high temperature solder, or equivalent.
		Bush (comm. end housing)	1	
		O-ring (comm. end housing)	1	
		O-ring (terminal)	2	
		Brush Assembly	1	
		O-ring (casing screws)	3	
		Snap Ring (nose housing - spares)	1	
		Pinion Stop (nose housing - spares)	1	
3806190	Alternator	Ball Bearing (drive end)	1	Reference Section 5 in this service bulletin (containing form 4626-1187) before commencing Alternator service or repair.
SK32	4626JC	Oil Seal (drive end)	1	
		Slip Ring	1	
		Roller Bearing (slip ring end)	1	
		Brush	2	
		Oil Seal (slip ring end housing)	1	
		Inner Bearing (slip ring end)	1	
		Oil Seal (slip ring end housing)	1	
		O-ring (terminal stud)	1	

3806191	Alternator	Bearing (front)	1	Reference Section 5 in this service bulletin (containing 125-121) before commencing Alternator service or repair
SK33	8SC30140	Bearing (rear housing)	1	
	8SC3009ZA	Oil Seal (rear housing)	2	
		Slip Ring	1	
		Brush Set	1	
		Bearing Seal (drive end)	2	
		Bearing (inner race)	1	
3806192	Alternator	Bearing (drive end)	1	Reference Section 5 in this service bulletin (containing 091996) before commencing Alternator service or repair. Apply Loctite™ grade 290, or equivalent, to the slip ring bore after assembly of the slip ring to rotor shaft (allow 12 hours cure time at room temperature (20°C)[68°F])
SK36	LNA 4024	Bearing (slip ring end)	1	
	LNA 5524	Brush Holder Assembly	1	
		Slip Ring (machined)	1	
		O-ring Seal (rear housing/stator)	1	
		Gasket (regulator assembly)	1	
3806198	Starter Motor	Brush Set (includes 8 brushes)	1	Reference Section 5 in this service bulletin (containing 4352-483) before commencing Starter motor service or repair.
SK37	M001 7406MC	O-ring (field coil housing)	2	
	M001 7288MC	Boot (lever arm)	1	
		Washer (drive pinion)	1	

		Wick (nose housing)	1	
		Thrust Washer (nose housing)	1	
		Seal Kit (shift housing)	1	
		Wick (comm. end housing)	1	
		Bush	1	
		O-ring (jumper terminal)	1	
		Bushing (insulation)	2	
		Bushing (insulation)	2	
		Bushing (insulation)	1	
		Cam (shift arm lever)	1	
3806193	Alternator	Bearing (drive end)	1	Reference Section 5 in this service bulletin (containing 3429-1071) before commencing Alternator service or repair. The slip ring is mounted inboard on the rotor, therefore a bearing change does not require removal of the slip ring. Replaces kit number 50051.
SK38	2300JB	Bearing (slip ring end)	1	
	2301JB	Brush	2	
	2303JB	Regulator Gasket (flat top)	1	
	2502JB	Gasket (regulator/slip ring end housing)	1	
		Regulator Gasket (dome top)	1	

3806194	Alternator	Teflon™ Lip Seal (fan end)	2	Reference Section 5 in this service bulletin (containing form 3191-469) before commencing Alternator service or repair.
SK39	2184AC	O-ring (fan end)	2	
		Ball Bearing (fan end)	1	
		O-ring (fan end)	1	
		Brush	2	
		O-ring (brush holder)	2	
		O-ring (brush holder)	2	
		Teflon™ Lip Seal (drive end)	1	
		Ball Bearing (drive end)	1	
		O-ring (drive end)	1	
		Teflon™ Seal (drive end)	1	
3806195	Alternator	Bearing and Seal Pack (slip ring end)	1	Reference Section 5 in this service bulletin (containing 3666624) before commencing Alternator service or repair
SK40	AMA 5104UT	Bearing and Seal Pack (drive end)	1	
		Brush Set	1	
		Brush Spring Set	1	
		Nut, Washer, Key pack (drive end)	1	

3806199	Starter Motor	Cam (shift arm lever)	2	Reference Section 5 in this service bulletin (containing 4352-483) before commencing Starter motor service or repair.
SK41	M001 094134	Bushing (nose & comm. end housing)	2	
		Bush (shift housing)	1	
		Wick (nose housing)	1	
		Wick (comm. end housing)	1	
		O-ring	1	
		O-ring (field coil housing)	2	
		O-ring (nose housing)	1	
		Brush Set (includes 8 brushes)	1	
		Thrust Washer (nose housing)	1	
		Seal (shift housing and field coil)	1	

Section 4. Warranty Information

The following is for Cummins® Branded Presolite™ Starters and Alternators Service and Warranty.

All Cummins Inc. engine plants phased in the introduction of Branded Prestolite™ Starters and Alternators fitted to engines from September 1997. These Starters and Alternators will have new dataplates which state that this product is built for Cummins Inc. by the accessory supplier.

These products will be covered under New Engine Warranty on all applications except for the U.S./Canada Recreational Vehicle (RV), School Bus, Fire and Crash Truck, where the coverage is limited to 2 years. The appropriate legal description has been revised to reflect this change of coverage. This warranty applies to the Cummins® Branded Starters and Alternators **only**. If a Cummins® branded Starter or Alternator fails during the warranty period due to a defect in material or workmanship, Cummins Inc., at its option, will replace/repair the Starter or Alternator without charge to the owner.

In order to receive warranty reimbursement for Cummins® branded Starters and Alternators, the Cummins Inc. branded dataplates **must** be confirmed.

The use of ReCon® is the standard method for replacing failed components. The warranty tag in the ReCon® box is to be completed, attached to the failed core and returned to Cummins Inc. UKDRC for the UK and MPDC for Europe, Middle East, and Africa. The warranty tag has to match with the submitted warranty claim. Repairs, instead of ReCon® exchange, could possibly be carried out if the cost is less, or if usage of ReCon®, is **not** feasible. For this, Cummins Inc. issued Rotary Electric repair kits have to be used in connection with proper tooling and by trained personnel. The Standard Repair Times for Starters and Alternators have been updated to reflect the use of these products.

Note : This is the sole and exclusive warranty applicable to these products. There are no other warranties, expressed or implied, including no warranties of fitness for a particular purpose and merchantability. Under no circumstances will Cummins Inc. be liable for incidental or consequential damage.

The following is for Cummins Inc. Warranty Guidelines.

The warranty of the Cummins Inc. Branded Starters and Alternators will be administered according to the guidelines in the Cummins Inc. Warranty Administration Manual, Bulletin 4021290, in reference to Cummins Inc. Form 6299.

Starter and Alternator malfunctions that are **not** considered warrantable are:

Standard Repair Times

Standard Repair Times - Starters		
Procedure Number	Description	Hours
13-603	Replace solenoid switch (includes check end play, adjust pinion position and test operation.)	0.5
13-604	Replace drive (includes clean, lubricate, check end play, adjust pinion position and test operation.)	0.5
13-605	Recondition (includes disassembly, cleaning, replace necessary parts, lubricate, seat brushes, assembly, and test operation.)	2.5
Standard Repair Times - Alternators		
Procedure Number	Description	Hours
13-126	Inspect and replace brushes, regulator, or external diode trio.	0.5

13-127	Recondition (includes disassembly, cleaning and testing all parts, replace as needed. Assembly and test)	1.5

Section 5. Service Literature

Please contact the manufacturer for additional service literature.

This Service Bulletin is intended to be used with:

Service Information Binder - Branded Rotary Electric (Prestolite™/Bosch™) 3666625.

Warranty Adjudication Posters (Alternators) 3666637.

Warranty Adjudication Posters (Starters) 3666638.

Service/Parts Topic 97T13-2.

Service Bulletin 3666191 (/qs3/pubsys2/xml/en/bulletin/3666191.html).

Section 6. Service Tools

This section outlines the tools required to troubleshoot and diagnose Starter and Alternator malfunctions. It is divided into three categories. The first category is for tools that are required for on-engine diagnostics. The second category is for over the counter exchanges and rebuilding. The third category covers Prestolite™ specific service tools.

Category 1 - Required On-Engine Diagnostics

On-Engine Diagnostics	
Description	Service Tool Part Number
Multimeter	3377161
Clamp on Current Probe	3823574
Carbon Pile Load Tester	3885847
Digital Optical Tachometer	3377462

Category 2 - Over the Counter Exchange or Rebuild

Counter Exchange or Rebuild	
Description	Service Tool Part Number
Bench Tester, 3 phase 400 VAC, 50 Hz	3825047
Bench Tester, 3 phase 440 VAC, 60 Hz	3825048

Category 3 - Prestolite™ Service Tools

Description	Service Tool Part Number
Tee Wrench	3824866
Tee Wrench	3824867

Timing Key	3824668
Timing Gauge	3824869
Abutment Gauge	3824870
LNS Starter Motor Abutment Gauge	3825060
MS1 Starter Motor Abutment Gauge	3825061
MS1 Starter Motor Timing Gauge	3825062

Section 7. Service Training

Service Training relevant to testing and troubleshooting the Cummins® Branded Prestolite™ Starters and Alternators is available. Details can be obtained from:

**Service Training Centre,
Customer Support Group
Cummins Inc. Ltd.
Royal Oak Way South,
Daventry, Northants, NN11 5NU.
ENGLAND
Telephone: (44) 01327 886000
Facsimile: (44) 01327 886118**

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